

ENTERPRISE BUSINESS REQUIREMENTS DOCUMENTATION

AI-Powered Healthcare Medical Institution Platform

Business Requirements Document

Volume IV of IV | Data Architecture, Analytics & Digital Experience Design

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This document and its contents are confidential and proprietary. It is intended solely for authorized internal review and downstream architecture, design, and engineering planning. All patient records referenced within are fictional. This document does not itself constitute application source code or a certification of regulatory compliance.

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Document Revision History

VERSION	DATE	AUTHOR	DESCRIPTION
1.0	September 2026	Enterprise Architecture & Digital Health Strategy	Initial draft issued for executive review.

32. Complete Patient Data Model

The data model below consolidates the required fields from the project specification with additional attributes necessary for a coherent patient record. Every field is classified into one of seven categories, which determines the handling requirements defined in Sections 9–10.

32.1 Classification Legend

CLASSIFICATION	MEANING
General	Non-sensitive descriptive information.
PII	Personally identifiable information requiring restricted access and encryption.
PHI	Protected health information requiring clinical-role access and full audit trail.
Clinical	Clinical/diagnostic content authored or reviewed by qualified staff.
Operational	Supports institutional workflow and throughput, not directly diagnostic.
Audit	System-generated record of an action, retained for accountability.
Analytics	Derived/aggregate field used for reporting and dashboards.

32.2 Patient Data Model Fields

FIELD	DESCRIPTION	CLASSIFICATION
Patient ID	Unique system identifier for the patient record.	General
Patient Full Name	Patient's legal name.	PII
Date of Birth	Used for identification and age-based clinical logic.	PII
Gender	Self-identified or recorded gender.	PII
Contact Information	Phone, email, and mailing address.	PII
Government/National ID Reference	Identifier used for identity verification.	PII
Patient General Information	Non-sensitive profile data (preferred language, communication preference).	General
Patient PII Information	Composite reference to all direct-identifier fields above.	PII
Medical Diagnosis	Current and historical diagnosis entries.	PHI
Patient Monitoring Details	Current monitoring parameters and status.	PHI
Patient Monitoring Log Date	Timestamp of each monitoring entry.	PHI
Patient Severity Status	Current severity classification (Critical / High-Risk / Moderate-Risk / Stable).	Clinical
Patient Health Score	Composite clinical indicator (0–100 scale).	Clinical
Medications Offered	Current and historical medication records.	PHI
Patient Next of Kin	Emergency contact name and relationship.	PII
Assigned Physician	Physician(s) of record for the patient.	Clinical
Patient Check-In Type	Emergency, Scheduled, Walk-In, or Follow-Up.	Operational

FIELD	DESCRIPTION	CLASSIFICATION
Number of Check-Ins	Total count of check-in events for the patient.	Analytics
Historical Check-In Dates	Full list of prior check-in timestamps.	Operational
Historical Checkout Dates	Full list of prior checkout timestamps.	Operational
Check-In Reasons	Reason recorded for each check-in event.	Clinical
Checkout Reasons	Reason/disposition recorded for each checkout event.	Clinical
Date of First Check-In	Earliest recorded check-in for the patient.	Analytics
Date of Last Check-In	Most recent recorded check-in.	Analytics
Last Patient Checkout Date	Most recent checkout timestamp.	Operational
Last Discharge Report	Summary content of the most recent discharge.	PHI
Admission History	Count and dates of inpatient admissions.	PHI
Emergency Room Visit Count	Total count of ER visits.	Analytics
Record Created Timestamp	When the patient record was created.	Audit
Record Last Modified Timestamp	When the patient record was last modified, and by whom.	Audit
Consent Status	Patient consent status for data use and AI-assisted educational content.	General
AI Content Association	References to AI-generated educational content linked to the patient, with review status.	Analytics

33. Synthetic Demonstration Patient Records (20 Records)

SYNTHETIC / DEMONSTRATION DATA — NOT REAL PATIENT INFORMATION

The 20 records below are entirely fictional and created solely to demonstrate the data model and to seed the analytics and dashboard examples in Sections 34–35. No real patient, living or deceased, is represented. Every synthetic record and every screen that displays it must carry a persistent, visible label identifying it as demonstration data.

33.1 Requirements

- Exactly 20 synthetic patient records are created for demonstration purposes.
- Every record and every UI surface displaying these records carries the label "SYNTHETIC / DEMONSTRATION DATA — NOT REAL PATIENT INFORMATION" (Section 33 banner) persistently, not just on first load.
- Records vary across diagnosis, severity, health score, check-in type, monitoring status, medications, admission history, discharge history, visit count, ER visits, and check-in/checkout reasons, per Section 33.2–33.4.
- Synthetic records are stored and access-controlled using the same classification and RBAC rules as real records (Sections 10, 11) so that the demonstration environment exercises genuine security behavior.

- Synthetic records are excluded from any production analytics/reporting once real patient data is in use, to prevent skewing operational metrics.

33.2 Identity & Check-In Summary (Records 1–20)

PATIENT ID	FICTIONAL NAME	AGE/SEX	CHECK-IN TYPE	# CHECK-INS	FIRST CHECK-IN	LAST CHECK-IN
SYN-P-001	Ava Thompson	34 / F	Emergency	5	2024-01-12	2026-08-02
SYN-P-002	Marcus Webb	58 / M	Scheduled	12	2019-03-04	2026-07-15
SYN-P-003	Priya Nair	27 / F	Walk-In	3	2025-11-02	2026-06-20
SYN-P-004	David Chen	71 / M	Emergency	8	2021-05-19	2026-08-25
SYN-P-005	Elena Rodriguez	45 / F	Follow-Up	6	2023-02-10	2026-08-28
SYN-P-006	James O'Brien	62 / M	Emergency	4	2024-09-01	2026-07-10
SYN-P-007	Grace Kim	8 / F	Walk-In	2	2026-01-15	2026-05-22
SYN-P-008	Robert Nguyen	39 / M	Emergency	7	2022-06-12	2026-08-30
SYN-P-009	Linda Park	55 / F	Scheduled	15	2018-04-22	2026-08-20
SYN-P-010	Samuel Osei	66 / M	Emergency	9	2020-10-05	2026-08-27
SYN-P-011	Nora Fitzgerald	29 / F	Walk-In	3	2025-03-18	2026-07-02
SYN-P-012	Hassan Ali	50 / M	Scheduled	11	2019-08-14	2026-08-05
SYN-P-013	Chloe Martin	19 / F	Emergency	2	2026-02-11	2026-06-30
SYN-P-014	Benjamin Wright	74 / M	Emergency	10	2019-01-09	2026-08-29
SYN-P-015	Aisha Rahman	41 / F	Follow-Up	5	2023-12-01	2026-08-10
SYN-P-016	Thomas Baker	5 / M	Emergency	2	2026-04-02	2026-08-18
SYN-P-017	Isabella Rossi	33 / F	Scheduled	4	2024-05-25	2026-07-28
SYN-P-018	Victor Ivanov	60 / M	Emergency	6	2022-02-17	2026-08-22
SYN-P-019	Sofia Mendes	48 / F	Walk-In	3	2025-01-30	2026-06-15
SYN-P-020	Kevin O'Malley	53 / M	Emergency	13	2017-11-11	2026-08-31

SYNTHETIC / DEMONSTRATION DATA — NOT REAL PATIENT INFORMATION

33.3 Clinical Snapshot (Records 1–20)

PATIENT ID	DIAGNOSIS	SEVERITY	HEALTH SCORE	MONITORING	MEDICATIONS
SYN-P-001	Acute Appendicitis	Critical	42	Active	IV Antibiotics, Analgesics
SYN-P-002	Type 2 Diabetes (routine)	Stable	78	Routine	Metformin
SYN-P-003	Migraine	Moderate-Risk	68	Routine	Sumatriptan
SYN-P-004	Congestive Heart Failure	High-Risk	51	Active	Furosemide, Lisinopril
SYN-P-005	Breast Cancer (post-treatment)	Moderate-Risk	64	Active	Tamoxifen
SYN-P-006	Ischemic Stroke	Critical	38	Active	Aspirin, Atorvastatin
SYN-P-007	Seasonal Allergies	Stable	88	Routine	Cetirizine
SYN-P-008	Type 1 Diabetes (DKA episode)	High-Risk	55	Active	Insulin (IV then SC)
SYN-P-009	Hypertension	Stable	74	Routine	Amlodipine

PATIENT ID	DIAGNOSIS	SEVERITY	HEALTH SCORE	MONITORING	MEDICATIONS
SYN-P-010	Chronic Kidney Disease (Stage 4)	Critical	33	Active	Erythropoietin, Phosphate Binders
SYN-P-011	Anxiety Disorder	Moderate-Risk	70	Routine	Sertraline
SYN-P-012	Coronary Artery Disease (post-stent)	High-Risk	58	Active	Clopidogrel, Atorvastatin
SYN-P-013	Fracture (Radius)	Moderate-Risk	72	Routine	Ibuprofen
SYN-P-014	COPD Exacerbation	Critical	40	Active	Albuterol, Prednisone, Oxygen
SYN-P-015	Rheumatoid Arthritis	Moderate-Risk	66	Routine	Methotrexate
SYN-P-016	Febrile Seizure (pediatric)	High-Risk	60	Active	Acetaminophen
SYN-P-017	Prenatal Care (3rd trimester)	Stable	84	Routine	Prenatal Vitamins
SYN-P-018	Sepsis (resolved)	Critical	45	Active	IV Antibiotics (broad-spectrum)
SYN-P-019	Hypothyroidism	Stable	76	Routine	Levothyroxine
SYN-P-020	Alcohol Withdrawal Syndrome	High-Risk	48	Active	Lorazepam (taper), Thiamine

SYNTHETIC / DEMONSTRATION DATA — NOT REAL PATIENT INFORMATION

33.4 Visit, Admission & Discharge History (Records 1–20)

PATIENT ID	ADMISSIONS	ER VISITS	LAST DISCHARGE REPORT	CHECK-IN REASON	CHECKOUT REASON
SYN-P-001	1 (current)	2	N/A — currently admitted	Severe abdominal pain	N/A (in-patient)
SYN-P-002	0	0	Routine follow-up completed; no acute findings	Quarterly diabetes management	Visit completed
SYN-P-003	0	1	Symptoms resolved with treatment	Severe headache	Symptom resolution
SYN-P-004	3	4	Discharged with adjusted diuretic regimen	Shortness of breath	Stabilized, discharged home
SYN-P-005	1 (past)	0	Oncology follow-up, stable	Oncology follow-up	Visit completed
SYN-P-006	2	2	N/A — currently admitted	Sudden weakness / slurred speech	N/A (in-patient)
SYN-P-007	0	0	Symptoms managed	Allergy symptoms	Visit completed
SYN-P-008	2	3	DKA resolved; education provided	Elevated glucose, nausea	Stabilized, discharged home
SYN-P-009	0	0	Blood pressure well controlled	Routine hypertension check	Visit completed
SYN-P-010	4	3	N/A — currently admitted	Fatigue, fluid overload	N/A (in-patient)
SYN-P-011	0	0	Referred to outpatient counseling	Anxiety symptoms	Visit completed
SYN-P-012	2	1	Cardiology follow-up stable	Cardiology follow-up	Visit completed
SYN-P-013	0	1	Cast applied; orthopedic follow-up scheduled	Fall injury	Treated and released
SYN-P-014	5	5	N/A — currently admitted	Severe breathlessness	N/A (in-patient)
SYN-P-015	0	0	Disease activity stable	Rheumatology follow-up	Visit completed
SYN-P-016	1	2	Observation completed; pediatric neuro follow-up advised	High fever with seizure	Stabilized, discharged home
SYN-P-017	0	0	Prenatal check normal	Routine prenatal visit	Visit completed

PATIENT ID	ADMISSIONS	ER VISITS	LAST DISCHARGE REPORT	CHECK-IN REASON	CHECKOUT REASON
SYN-P-018	2	2	Sepsis resolved; discharged with oral antibiotics	High fever, confusion	Stabilized, discharged home
SYN-P-019	0	0	Thyroid levels within target	Routine thyroid check	Visit completed
SYN-P-020	3	6	Withdrawal managed; referred to substance use program	Tremors, agitation	Stabilized, discharged with referral

33.5 Distribution Summary (For Analytics Demonstration)

- Severity distribution: 5 Critical, 5 High-Risk, 5 Moderate-Risk, 5 Stable — an intentionally even spread to exercise every severity view in Section 34.1.
- Check-in type distribution: 9 Emergency, 4 Scheduled, 4 Walk-In, 2 Follow-Up (approximate), giving realistic variation for Section 34.6.
- Two pediatric records (SYN-P-007, SYN-P-016) and one prenatal-care record (SYN-P-017) are included to demonstrate age/condition diversity.
- Four records currently show an active in-patient admission with no checkout/discharge yet recorded (SYN-P-001, SYN-P-006, SYN-P-010, SYN-P-014), to demonstrate the admitted-patient dashboard state.

34. Healthcare Analytics Requirements

Each analytics area below specifies the metrics to track and the recommended visualization type, chosen for clarity over decoration per the design principles in Section 37.

34.1 Patient Severity

Metrics

Count and % of patients by severity (Critical, High-Risk, Moderate-Risk, Stable); severity trend over time.

Recommended Visualization

Donut/pie chart for current distribution; stacked area chart for trend over time.

Purpose

Gives clinical and administrative leadership an immediate read on institutional acuity and load.

34.2 Patient Health

Metrics

Average health score (institution-wide and by unit); health score distribution; monitoring trends per patient.

Recommended Visualization

KPI card for average; histogram/bar chart for distribution; line chart for individual/cohort trend.

Purpose

Surfaces population-level health trajectory and flags cohorts trending downward.

34.3 Diagnosis

Metrics

Diagnosis frequency; category breakdown; trend over time.

Recommended Visualization

Horizontal bar chart for frequency/category; line chart for trend.

Purpose

Identifies the institution's clinical caseload mix and shifts in demand.

34.4 Admissions

Metrics

Daily / weekly / monthly admission counts; trend.

Recommended Visualization

Line chart for trend; bar chart for daily/weekly/monthly comparison.

Purpose

Supports staffing and bed-capacity planning.

34.5 Discharges

Metrics

Discharge volume; average length of stay; trend.

Recommended Visualization

Bar chart for volume; line chart for average length of stay over time.

Purpose

Tracks throughput efficiency and bed turnover.

34.6 Check-Ins

Metrics

Volume; type breakdown (Emergency/Scheduled/Walk-In/Follow-Up); reasons; historical patterns.

Recommended Visualization

Bar chart for volume/type; table for top reasons; line chart for historical pattern.

Purpose

Reveals front-desk and triage load and its underlying causes.

34.7 Medications

Metrics

Utilization counts; category breakdown; trend.

Recommended Visualization

Bar chart for utilization/category; line chart for trend.

Purpose

Supports medication safety review and pharmacy/operational planning.

34.8 Emergency Room

Metrics

ER visit volume; severity mix of ER visits; trend.

Recommended Visualization

KPI card for current volume; stacked bar chart for severity mix; line chart for trend.

Purpose

Directly informs ER staffing and resource allocation.

34.9 Healthcare Operations

Metrics

Patient throughput; operational efficiency indicators; workload; resource utilization.

Recommended Visualization

KPI cards for headline indicators; bar chart for workload by unit; gauge/trend indicator for utilization.

Purpose

Gives administrators a real-time operational efficiency view.

34.10 Patient Monitoring

Metrics

Monitoring frequency; severity trends; health-score trends; escalation indicators.

Recommended Visualization

Line chart for frequency/trend; trend indicator (up/down/stable arrow) for escalation.

Purpose

Surfaces patients whose condition is trending toward escalation before a crisis point.

35. Executive Dashboard

The Executive Dashboard consolidates the highest-priority indicators for administrators and executives into a single, role-scoped view (per RBAC, Section 11), combining aggregate metrics rather than individual patient detail.

DASHBOARD ELEMENT	RECOMMENDED VISUALIZATION	BUSINESS / HEALTHCARE PURPOSE
Total Patients	KPI Card	Baseline institutional population size for context on all other metrics.
Active Patients	KPI Card + trend indicator	Distinguishes current caseload from historical total; informs staffing.
Critical Patients	KPI Card (severity-colored) + trend indicator	Immediate visibility into the highest-acuity population requiring priority attention.
High-Severity Patients	KPI Card (severity-colored)	Complements Critical count to show the full high-acuity picture.
Average Health Score	KPI Card + trend indicator	Single-number population health signal for executive review.

DASHBOARD ELEMENT	RECOMMENDED VISUALIZATION	BUSINESS / HEALTHCARE PURPOSE
Today's Check-Ins	KPI Card	Real-time operational load indicator for front-desk/triage staffing.
Today's Admissions	KPI Card	Real-time inpatient demand signal for bed management.
Today's Discharges	KPI Card	Paired with admissions to show net inpatient census movement.
Emergency Room Activity	Bar chart (volume) + severity-colored breakdown	Highlights current ER load and acuity mix for resource allocation.
Medication Activity	Line chart (utilization over time)	Tracks medication administration volume for safety and operational review.
Monitoring Alerts	Table with severity indicators	Surfaces patients with escalating status requiring clinical attention.
AI-Generated Reports	Table with validation/approval status indicators	Gives executives visibility into AI-assisted report activity and its governance status (Section 36).

35.1 Visualization Type Guidance

- KPI Cards: single-number, at-a-glance metrics (totals, counts, averages) with an optional trend indicator (▲/▼/—) for immediate context.
- Line Charts: time-series trends (admissions, discharges, medication utilization, health-score trend) where direction over time matters most.
- Bar Charts: comparisons across discrete categories (diagnosis frequency, check-in type, ER volume by day).
- Pie/Donut Charts: proportional composition of a whole, used sparingly — primarily for severity distribution where the "share of total" framing is meaningful.
- Area Charts: cumulative or stacked trends (e.g., severity mix over time) where both total volume and composition matter simultaneously.
- Tables: precise, scannable detail (monitoring alerts, AI report status) where exact values and multiple attributes per row matter more than shape.
- Trend Indicators: compact up/down/stable markers attached to KPI cards to convey direction without a full chart.
- Severity Indicators: consistent color coding (Section 38) applied across every visualization type so severity is recognizable at a glance regardless of chart form.

36. AI-Generated Reporting Requirements

AI-generated reports accelerate drafting but remain subject to the full validation and approval pipeline defined in Section 16. Every report type below produces a draft, never a final publication, until reviewed.

36.1 Report Types

- Patient Population Summaries
- Operational Summaries
- Admission Summaries
- Discharge Summaries
- Severity Summaries

- Monitoring Summaries
- Analytics Summaries
- Educational Reports

36.2 Mandatory Report Metadata

METADATA FIELD	REQUIREMENT
AI-Generated Designation	A persistent, visible label identifying the report as AI-generated (Section 14.1).
Source / Context	The data range, patient cohort, or knowledge source the report draws from, where applicable.
Validation Status	Result of automated safety/accuracy validation (Section 16.1, Stages 3–4).
Approval Status	Pending / Approved / Rejected / Revision Requested, reflecting the Section 16.1 workflow outcome.
Timestamp	Generation timestamp and, separately, the approval/rejection timestamp.
Version	Version identifier linking this report to its draft history (Section 16.3).
Responsible Reviewer	Name/role of the human reviewer who approved, rejected, or requested revision, where applicable.
Audit Record	Reference to the corresponding audit log entries (Section 12) covering generation and approval events.

36.3 Requirement Statement

No AI-generated report — regardless of type — is considered final or distributable until it carries an Approved status from a qualified reviewer per Section 16.1. Reports pending review are visually distinguished (e.g., a "Draft — Pending Review" state) from approved reports throughout the Reporting page and Executive Dashboard (Section 35).

37. Website Design System

The design system establishes a premium, trust-oriented healthcare technology aesthetic that prioritizes usability and clarity over decorative effect, consistent with the site-wide standards introduced in Section 7.12.

37.1 Design Principles

- **Light Primary Theme:** content areas use a light, high-contrast background to maximize readability for clinical and administrative data.
- **Dark Hero Sections:** landing/entry moments (Home hero, Admin Login) use a dark, high-contrast treatment to establish a modern, premium first impression.
- **Modern, Trust-Oriented Appearance:** visual language favors clean geometry, generous whitespace, and restrained color use over ornamentation, signaling institutional credibility.
- **Premium Healthcare Aesthetic:** typography, spacing, and color are consistent with contemporary digital-health products rather than a generic corporate template.
- **Futuristic but Professional:** selective use of contemporary type and subtle motion (Section 39) suggests technological sophistication without compromising clinical seriousness.
- **Usability Over Decoration:** any interactive or motion effect must support comprehension (e.g., revealing a chart on scroll) — decoration that does not aid understanding is excluded.

38. Color & Typography System

38.1 Color System

ROLE	COLOR	USAGE	ACCESSIBILITY NOTE
Primary	Deep Navy Blue	Primary brand color; navigation, headings, primary buttons.	Meets WCAG AA against white/light backgrounds.
Secondary	Teal	Secondary accents, section headers, links.	Meets WCAG AA against white/light backgrounds.
Accent	Amber/Gold	Highlights, callouts, selective emphasis — used sparingly.	Reserved for non-critical emphasis; not used for status alone.
Background	Off-White / Light Neutral	Primary page background in content areas.	Maximizes contrast for body text and data.
Surface	White / Very Light Gray	Cards, tables, dashboard panels.	Subtle elevation from background without relying on color alone.
Primary Text	Near-Black / Deep Charcoal	Body copy, headings on light surfaces.	High contrast (7:1+) for extended reading of clinical content.
Muted Text	Mid Gray	Secondary labels, captions, timestamps.	Meets minimum 4.5:1 contrast on light surfaces.
Success	Green	Positive states (approved, stable, resolved).	Paired with icon/label, not color alone, for colorblind accessibility.
Warning	Amber	Caution states (pending review, moderate risk).	Paired with icon/label, not color alone.
Error	Red	Failed actions, access denied, rejected content.	Paired with icon/label, not color alone.
Critical Severity	Deep Red / Crimson	Reserved exclusively for Critical patient severity indicators.	Visually distinct from standard Error red to avoid confusing system errors with patient acuity.

38.2 Accessibility Contrast Requirement

All text/background pairings meet WCAG 2.1 AA contrast minimums (4.5:1 for body text, 3:1 for large text/UI components). Severity and status information is never conveyed by color alone — every color-coded indicator is paired with a text label or icon (Section 38.1, Success/Warning/Error/Critical rows).

38.3 Typography System

CONTEXT	TYPOGRAPHY TREATMENT	RATIONALE
Headings (H1–H2)	Contemporary geometric sans-serif, bold weight	Large scale for section/page titles; establishes visual hierarchy.
Headings (H3+) / Subheadings	Same family, medium-bold weight, smaller scale	Maintains hierarchy without competing with H1/H2.
Body Text	Highly legible humanist sans-serif	Optimized for extended reading of clinical and administrative content.
Navigation	Same body family, medium weight, slightly condensed	Compact but legible for menus and wayfinding.
Dashboards / KPI Cards	Tabular/numeric-optimized sans-serif for figures; body family for labels	Numbers align cleanly in cards and tables; labels stay readable at small size.

CONTEXT	TYPOGRAPHY TREATMENT	RATIONALE
Selected Futuristic Headings (Home Hero)	Contemporary display sans-serif with wider letterforms, used only for the hero headline	Delivers the modern/futuristic impression in Section 10 without compromising body-text readability elsewhere.

39. User Experience Requirements

39.1 Responsive Design

- Desktop: full multi-column layouts for dashboards and data tables.
- Tablet: collapsible navigation and reflowed 2-column layouts.
- Mobile: single-column stacking, touch-optimized controls, and summarized (not truncated) data views.

39.2 Interactive Components & Scroll Effects

- Interactive elements (expandable sections, chart tooltips, filters) support comprehension, not decoration (Section 37.1).
- Scroll effects are limited to purposeful reveal of content (e.g., staged reveal of Home page sections) and never delay access to critical information.

39.3 Navigation

- Persistent, role-aware primary navigation reflecting the user's access level (Section 11).
- Clear indication of current location within the site/app (breadcrumbs or active-state highlighting).

39.4 Accessibility

- WCAG 2.1 AA conformance target across all patient-facing and clinical interfaces (Section 6, NFR-Accessibility).
- Full keyboard navigability and screen-reader compatibility, including AI narration integration (Section 17.3).

39.5 Forms

- Inline validation with clear, specific error messaging.
- Minimum-necessary field collection consistent with Section 10.1 data minimization.

39.6 Data Tables

- Sortable/filterable columns for administrative and analytics tables.
- Sticky headers for long patient/record lists; pagination rather than infinite unbounded scroll for record sets.

39.7 Dashboards

- Role-scoped default views (Section 11) with consistent KPI card and chart placement (Section 35).
- Configurable widget layout where appropriate for analyst/executive roles.

39.8 Patient Records

- Clear visual separation between General/PII, PHI, and Clinical sections within a single patient view (Section 32.1 classification reflected in UI grouping).
- Persistent severity indicator visible wherever a patient record is displayed.

39.9 AI Interfaces

- Visible AI-generated labeling and validation/approval status on every AI output (Sections 14.1, 36.2).
- Clear differentiation between "Draft — Pending Review" and "Approved" content states.

39.10 Error Handling

- Specific, actionable error messages (never generic "something went wrong" for user-correctable errors).
- Security-sensitive errors (e.g., failed authentication) avoid revealing whether an account exists.

39.11 Loading States

- Skeleton/placeholder loading for dashboards and data tables rather than blank screens.
- Progress indication for AI generation requests (Section 16.1), reflecting pipeline stage where feasible.

39.12 Empty States

- Meaningful empty-state messaging (e.g., "No monitoring alerts at this time") rather than a blank panel.
- Empty states for new/demo environments clearly distinguish "no data yet" from "data unavailable due to an error."

39.13 Confirmation Dialogs

- Explicit confirmation required for destructive or high-impact actions (record deletion, discharge approval, AI content rejection).
- Confirmation dialogs state the specific action and object being affected, not a generic "Are you sure?"

39.14 Administrative Workflows

- Visual progress indication through multi-step administrative workflows (e.g., approval pipelines, Section 24).
- Clear role-based visibility into what stage a pending action is at and who is responsible next.

40. Page Design Specifications

Each page specification below covers Purpose, Layout, Sections, Components, User Interactions, Data Displayed, Security Considerations, and Responsive Behavior, extending the page-level requirements introduced in Section 7.

40.1 Home

Purpose

Introduce the institution and platform; orient visitors toward the right next step.

Layout

Full-width dark hero (Section 41) followed by light-theme content sections in a single scrolling column.

Sections

Hero; Value Proposition; Services Highlight; Responsible AI Messaging; Trust Indicators; Call-to-Action Footer.

Components

Primary/secondary CTA buttons, service highlight cards, trust-indicator badges, scroll-reveal sections.

User Interactions

CTA navigation to login/registration; scroll-triggered section reveal; hover states on highlight cards.

Data Displayed

Static/CMS marketing content only.

Security Considerations

No PII/PHI; publicly accessible.

Responsive Behavior

Hero headline and CTA stack vertically on mobile; highlight cards reflow from 3-column to 1-column.

40.2 About Us

Purpose

Communicate institutional background, mission, and credibility.

Layout

Single-column light-theme long-form content with supporting imagery blocks.

Sections

Mission/Vision; Institutional History; Accreditation/Credibility Indicators; Leadership Overview (optional).

Components

Text blocks, credibility badges, optional team/leadership cards.

User Interactions

Minimal — primarily reading; optional expandable sections for leadership bios.

Data Displayed

Static/CMS-managed content.

Security Considerations

No PII/PHI; publicly accessible.

Responsive Behavior

Image/text blocks stack vertically on tablet/mobile.

40.3 Contact Us

Purpose

Provide a channel for inquiries and support requests.

Layout

Two-column desktop layout (form + contact details), single column on mobile.

Sections

Contact Form; Institutional Contact Details; Location/Hours.

Components

Validated form fields, submit button, confirmation state, map/location reference.

User Interactions

Inline form validation; submission with visible confirmation state.

Data Displayed

Inquiry records (non-clinical), routed to staff.

Security Considerations

Rate-limited submissions; no sensitive health data collected via this form.

Responsive Behavior

Form and contact details stack vertically on mobile; touch-friendly input sizing.

40.4 Services

Purpose

Describe healthcare services and platform capabilities.

Layout

Categorized grid/list of service offerings with expandable detail.

Sections

Service Categories; Department Overviews; Platform Capability Highlights.

Components

Category cards, expandable detail panels, icons per service line.

User Interactions

Expand/collapse service detail; category filtering (optional).

Data Displayed

Static/CMS-managed content.

Security Considerations

No PII/PHI; publicly accessible.

Responsive Behavior

Grid reflows from multi-column to single column; expandable panels remain touch-accessible.

40.5 Patient Data

Purpose

Provide authorized users a secure, classified view of an individual patient record.

Layout

Sectioned single-patient view: header (identity + severity indicator) followed by classified data groupings.

Sections

Identity/PII (collapsed by default); PHI/Clinical (diagnosis, medication, monitoring); History (check-in/checkout, admissions); Next of Kin.

Components

Persistent severity badge, classification-grouped panels, monitoring trend mini-chart, audit-visible edit controls.

User Interactions

Role-gated edit actions; expand/collapse classified sections; navigate to full monitoring/analytics detail.

Data Displayed

Full patient record per Section 32, rendered per the requesting role's RBAC scope (Section 11).

Security Considerations

Strict RBAC; every view/edit logged (Section 12); session timeout enforced.

Responsive Behavior

Classified sections stack vertically on mobile; severity badge remains persistently visible while scrolling.

40.6 Dashboard

Purpose

Provide role-appropriate operational and clinical visualizations.

Layout

Grid of KPI cards at top, followed by chart/table panels (Section 35 layout).

Sections

KPI Summary Row; Severity & Health Panels; Operational Panels; Monitoring Alerts Table.

Components

KPI cards, line/bar/donut charts, alert table with severity indicators, date-range filter.

User Interactions

Date-range filtering; drill-down from KPI card to detail chart; hover tooltips on charts.

Data Displayed

Aggregated operational/clinical metrics per Section 34, scoped per role.

Security Considerations

Role-scoped data visibility; aggregate/de-identified views for broader analytics roles.

Responsive Behavior

KPI grid reflows from 4-column to 2-column to 1-column; charts resize/simplify on mobile.

40.7 Admin Login

Purpose

Provide secure authenticated entry for administrative and clinical staff.

Layout

Centered single-column form on a dark hero-style background (Section 37.1).

Sections

Login Form; MFA Step; Password Reset Link.

Components

Credential fields, MFA input, error messaging, password-reset flow entry point.

User Interactions

Submit credentials; complete MFA challenge; initiate password reset.

Data Displayed

Authentication records; login attempt logs (Section 12).

Security Considerations

MFA required; brute-force protection; generic error messaging that avoids confirming account existence.

Responsive Behavior

Form remains centered and touch-friendly at all breakpoints; MFA input optimized for mobile entry.

40.8 AI Features

Purpose

Provide governed access to AI capabilities (education, narration, image/audio generation, AI-assisted reporting requests).

Layout

Tabbed or card-based selection of AI capabilities, each opening a request/status panel.

Sections

Capability Selector; Request Form; Draft/Review Status; Approved Output Viewer.

Components

Capability cards, request form, status badges (Pending/Approved/Rejected), AI-generated content labels, narration player.

User Interactions

Submit AI request; view pipeline status (Section 16.1); play/view approved content.

Data Displayed

AI request/response records; governance review status (Section 36.2).

Security Considerations

All patient-facing/clinical-support AI content gated behind governance review; full request/response logging.

Responsive Behavior

Capability cards stack on mobile; narration player and status badges remain fully usable at small width.

40.9 Reporting

Purpose

Provide access to operational, clinical, and administrative reports, including AI-assisted drafts.

Layout

Filterable report list with a detail panel for the selected report.

Sections

Report Filters; Report List; Report Detail (with metadata per Section 36.2).

Components

Filter controls, report list table, metadata panel, export control (role-gated).

User Interactions

Filter/search reports; open report detail; request AI-assisted draft; export (where permitted).

Data Displayed

Aggregated/structured report data and AI report metadata (Section 36.2).

Security Considerations

Role-scoped report access; audit logging of generation and export (Section 12).

Responsive Behavior

List/detail layout collapses to a single navigable stack on mobile.

40.10 Admin Portal

Purpose

Central workspace for administrative account, role, and permission management.

Layout

Table-driven management interface with modal/panel-based edit flows.

Sections

Account Management; Role & Permission Management; Approval Workflow Queue.

Components

Data tables, role/permission editors, approval queue with action buttons, confirmation dialogs (Section 39.13).

User Interactions

Create/edit/delete accounts; assign roles/permissions; approve/reject pending items.

Data Displayed

Administrative account and configuration records.

Security Considerations

Highest-privilege access; MFA required; full audit logging of every action (Section 12).

Responsive Behavior

Tables switch to a stacked card view per record on mobile; critical actions remain reachable without horizontal scrolling.

40.11 Admin Dashboard

Purpose

Provide administrators with institution-wide operational and governance visibility.

Layout

Executive-style KPI and summary layout (Section 35), with an added governance/audit summary panel.

Sections

Operational KPI Row; Governance & Audit Summary; AI Governance Status Panel; System Health Indicators.

Components

KPI cards, audit summary table, AI governance status chart, system health indicators.

User Interactions

Drill down from summary panels into full audit/reporting detail (Section 40.9/40.10).

Data Displayed

Aggregated operational, audit, and AI governance data.

Security Considerations

Restricted to administrative roles; emphasizes aggregate/summary data over individual patient detail.

Responsive Behavior

Panel grid reflows to single-column on mobile; governance summary remains fully readable without horizontal scroll.

41. Home Page Hero

The Home page hero is the platform's first impression and must balance a modern, premium technology feel with immediate healthcare credibility.

REQUIREMENT	SPECIFICATION
Dark Visual Treatment	Deep navy/charcoal background with high-contrast light text, consistent with Section 37.1 and Section 38.1.
Strong Value Proposition	A single, clear headline stating what the platform does for patients and staff — not a generic tagline.
Modern/Futuristic Presentation	Selected futuristic heading typography (Section 38.3) and restrained motion/scroll accents, used only in this section.
Clear Call-to-Action	One primary CTA (e.g., "Get Started" / role-appropriate entry point) visually dominant over any secondary CTA.
Healthcare Imagery	Imagery or illustration that reads as clinical/technological rather than generic stock photography, reinforcing credibility.
Responsible AI Messaging	A concise statement that AI is used to support — not replace — clinical judgment, linking to further detail (Section 14–19) for interested visitors.
Professional Trust Indicators	Institutional credibility markers (e.g., accreditation references, security/privacy commitment statement) placed near the CTA to reduce visitor hesitation.

42. Design System Summary

DESIGN AREA	REQUIREMENT	UX OBJECTIVE	ACCESSIBILITY CONSIDERATION
Color System	Defined role-based palette (Primary, Secondary, Accent, Background, Surface, text, status colors) per Section 38.1.	Consistent, recognizable visual language across every page and role.	WCAG 2.1 AA contrast minimums; status never conveyed by color alone.

DESIGN AREA	REQUIREMENT	UX OBJECTIVE	ACCESSIBILITY CONSIDERATION
Typography System	Contemporary sans-serif system scaled for headings, body, navigation, dashboards, and a selective futuristic hero treatment (Section 38.3).	Readability under clinical/administrative workload; premium first impression on Home.	Legible at small sizes in tables/KPI cards; scalable for browser zoom/text-resize.
Layout & Theme	Light primary theme with dark hero sections (Section 37.1).	Balances premium first impression with sustained readability for data-heavy pages.	Sufficient contrast maintained in both light and dark treatments.
Responsive Behavior	Desktop/tablet/mobile breakpoints defined per component (Section 39.1).	Full functionality preserved across devices used by patients and staff.	Touch target sizing and reflow tested at each breakpoint.
Interactive & Scroll Effects	Purposeful, comprehension-supporting interactions only (Section 39.2).	Modern feel without sacrificing usability or clarity.	Effects do not trigger motion-sensitivity issues; reduced-motion preference respected.
Severity & Status Indicators	Consistent color + label/icon pairing across dashboards, patient records, and AI content (Sections 34–36, 38.1).	Instant recognizability of patient acuity and content status anywhere in the platform.	Distinguishable without color perception (icon/label always present).
AI Content Labeling	Persistent AI-generated/validation/approval labeling wherever AI content appears (Sections 14.1, 36.2, 39.9).	Transparency and trust in AI-assisted content.	Label text-based (not color/icon-only) so it is screen-reader accessible.
Forms & Data Entry	Inline validation, minimum-necessary fields, clear error states (Section 39.5, 39.10).	Reduces user error and administrative rework.	Errors announced to assistive technology; not conveyed by color alone.
Navigation	Role-aware, persistent primary navigation with clear current-location indication (Section 39.3).	Efficient wayfinding for both first-time and frequent users.	Fully keyboard-operable; visible focus states.

43. Final Requirements Traceability Matrix

This matrix traces each major requirement area of the original project specification across every layer of this BRD, confirming full coverage from business requirement through architecture, workflow, data, AI governance, UI, and analytics.

ASSIGNMENT REQUIREMENT	BRD SECTION	FUNCTIONAL REQ.	ARCHITECTURE COMPONENT	WORKFLOW	DATA REQUIREMENT	RESPONSIBLE AI CONTROL	UI / PAGE	ANALYTICS REQUIREMENT
Patient Management	Sec 5.1	FR-PAT-001–012	Application / Data Layer	Patient Workflow (22)	Patient Data Model (32)	N/A	Patient Data Page (40.5)	Check-In / Diagnosis Analytics (34.3, 34.6)
Administrative Management	Sec 5.2	FR-ADM-001–009	Application / Security Layer	Admin Workflow (24)	Admin Records (13.2)	N/A	Admin Portal (40.10)	Operational Metrics (34.9)
Healthcare Operations	Sec 5.3	FR-OPS-001–006	Application Layer	Hospital Workflow (23)	Operational Data (20.4)	N/A	Dashboard (40.6)	ER / Admission Analytics (34.4, 34.8)
Healthcare Analytics	Sec 5.4	FR-ANA-001–010	Application / Data Layer	N/A	Analytics Data (20.4)	N/A	Dashboard (40.6)	Sec 34 (all areas)
AI Educational Content	Sec 5.5	FR-AI-001	AI Layer	Responsible AI Workflow (26)	AI Knowledge Repository (29)	Sec 14–19	AI Features (40.8)	AI Content Engagement (36)
AI-Assisted Reporting	Sec 5.5	FR-AI-002	AI Layer / Application	Responsible AI Workflow (26)	Reporting Data (20.4)	Sec 14–19	Reporting (40.9)	AI Report Metadata (36.2)
Text-to-Image Generation	Sec 5.5, 17.1	FR-AI-003	AI Layer	Sec 17.1, 26	AI Knowledge Repository (29)	Sec 15, 17.1, 19	AI Features (40.8)	N/A
Text-to-Audio / Narration	Sec 5.5, 17.2–17.3	FR-AI-004/005	AI Layer	Sec 17.2–17.3, 26	N/A	Sec 14, 16, 19	AI Features (40.8)	N/A
Customized LLM	Sec 5.5, 10, 18	FR-AI-006	AI Layer (29)	Responsible AI Workflow (26)	Knowledge Repository (18.1)	Sec 18	AI Features (40.8)	N/A
HIPAA / Privacy Safeguards	Sec 9	NFR-Security/Privacy	Security Layer	RBAC Workflow (25)	PII/PHI Data (10.2, 32)	Sec 9	All Authenticated Pages	N/A
RBAC	Sec 11	NFR-Security	Security Layer	RBAC Workflow (25)	Admin Data	Sec 11	Admin Login / Portal (40.7, 40.10)	N/A
Audit Logging	Sec 12	NFR-Auditability	Security Layer	All Workflows (22–26)	Audit Logs (32.2)	Sec 12	Admin Dashboard (40.11)	N/A
Data Governance & Retention	Sec 13	NFR-Data Integrity	Data Layer	N/A	All Data Categories (32)	Sec 13	N/A	N/A
Responsible AI Governance	Sec 14–19	FR-AI-007/008	AI Layer	Responsible AI Workflow (26)	AI Content Records	Sec 14–19	AI Features (40.8)	AI Report Metadata (36.2)
Enterprise Architecture	Sec 20–21	N/A	All Layers	Enterprise Diagram (21)	N/A	Sec 19 (cross-cutting)	N/A	N/A
Customized LLM Architecture	Sec 29	FR-AI-006	AI Layer	Sec 29 diagram	Knowledge Repository (18.1)	Sec 18	AI Features (40.8)	N/A
Executive Dashboard	Sec 35	FR-ANA-009	Application / Data Layer	N/A	Analytics Data (20.4)	N/A	Dashboard (40.6)	Sec 35 (all KPIs)
Website Design	Sec 37–39	Sec 7 (Website	Presentation Layer	N/A	N/A	Sec 14.1 (AI	All Pages (40)	N/A

ASSIGNMENT REQUIREMENT	BRD SECTION	FUNCTIONAL REQ.	ARCHITECTURE COMPONENT	WORKFLOW	DATA REQUIREMENT	RESPONSIBLE AI CONTROL	UI / PAGE	ANALYTICS REQUIREMENT
System		Reqs)				labeling)		
Page Design Specifications	Sec 40	Sec 7	Presentation Layer	N/A	N/A	N/A	All 11 Pages (40.1–40.11)	N/A
Synthetic Demonstration Data	Sec 33	N/A	Data Layer	N/A	Patient Data Model (32)	N/A	Patient Data Page (demo)	All Analytics (demo, Sec 34)

Coverage check: every numbered element of the original four-volume project specification — vision, objectives, scope, stakeholders, functional and non-functional requirements, website requirements, HIPAA/privacy, PII/PHI, RBAC, audit logging, data governance, Responsible AI, AI safety, AI output validation, generative AI features, Customized LLM, AI governance, enterprise architecture, all required workflows, data flow, Replit integration, architectural principles, the architecture decision matrix, the patient data model, the 20 synthetic records, healthcare analytics, the executive dashboard, AI-generated reporting, and the full design system — is represented across Volumes I–IV and cross-referenced in the matrix above.

Final Document Approval

The complete four-volume Business Requirements Document is submitted for executive review and sign-off by the stakeholders below prior to advancement into detailed design and engineering.

ROLE	NAME	SIGNATURE	DATE
Executive Sponsor			
Chief Information Officer			
Chief Technology Officer			
Chief Information Security Officer			
Chief Medical Information Officer			

This concludes Volume IV and the complete Business Requirements Document. This document defines requirements, architecture, workflows, data models, synthetic demonstration data, analytics, and design-system specifications only. It is not a certification of regulatory (including HIPAA) compliance, and contains no application source code.